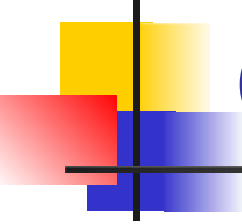
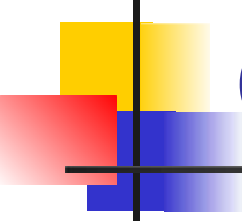


2G(GSM)



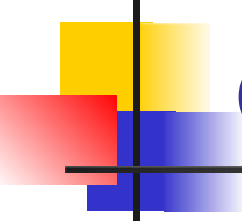
GSM

- Abbreviation for Global System for Mobile Communications
- Concurrent development in USA and Europe in the 1980's
- The European system was called GSM and deployed in the early 1990's



Goals of GSM

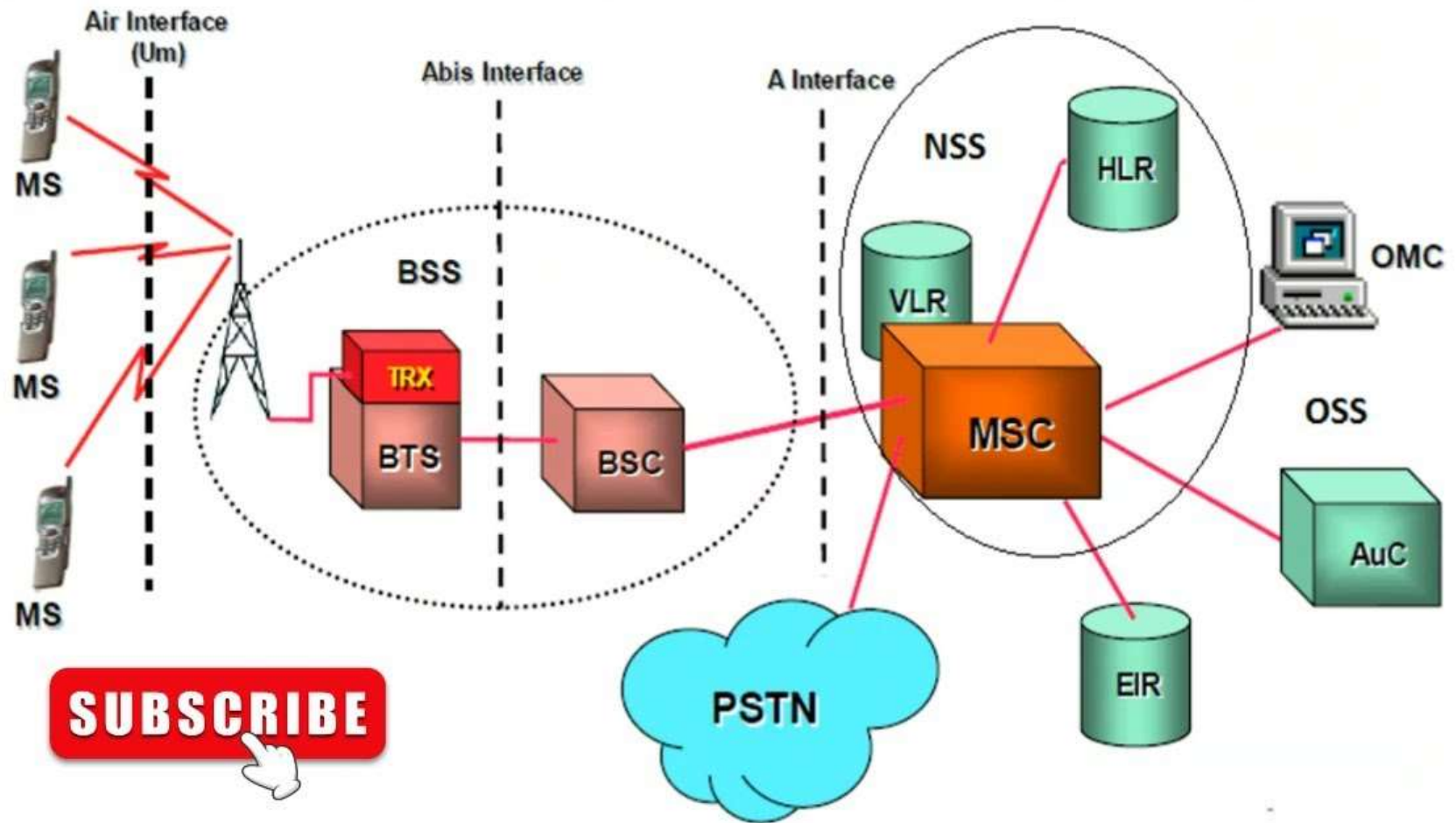
- Good subjective speech quality Low terminal
- and service cost Support for international
- roaming
- Ability to support handheld terminals
- Support for range of new services and facilities
- Spectral efficiency ISDN compatibility
-



GSM characteristics

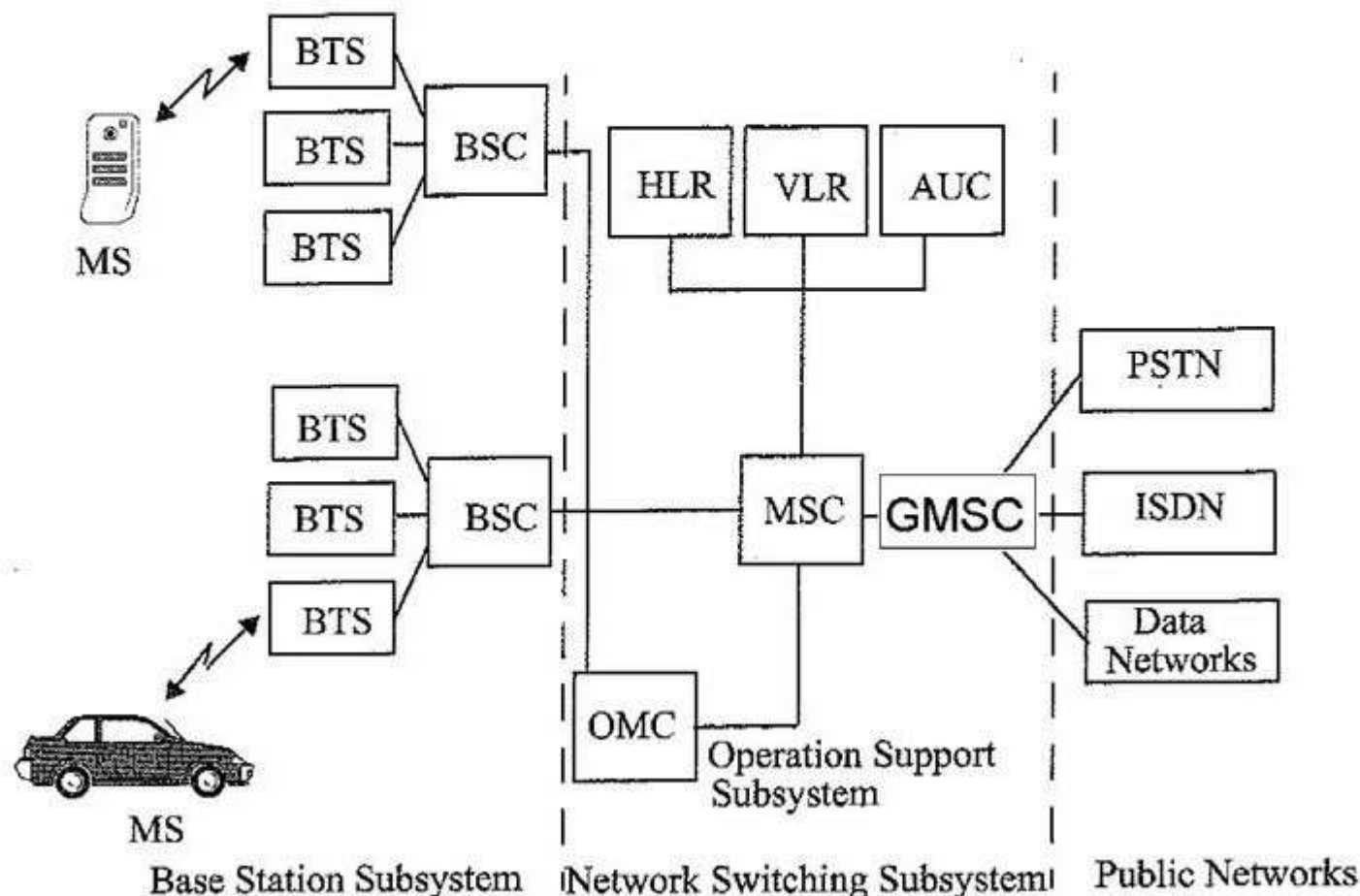
- previous standard in cellular communication were restrictive
- GSM – global digital standard for cellular phones that offered roaming facility
- first named Groupe Special Mobile and used in Europe; then usage extended to other continents
- GSM operate in frequency bands: 900MHz, 1800 MHz, 1900 MHz
- GSM provides voice and data services
-

GSM ARCHITECTURE



Global System for Mobile Communications

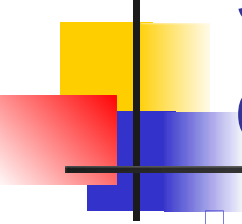
Basic architecture



Components of a cellular phone (MSU – Mobile Subscriber Unit)

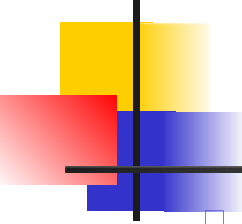
- radio transceiver – low power radio transmitter and receiver
- antenna, usually located inside the phone
- control circuitry – formats the data sent to and from the BTS; controls signal transmission and reception
- man-machine interface – consists from a keypad and a display; is managed by the control circuitry
- Subscriber Identity Module (SIM) – integrated circuit card that stores the identity information of subscriber
- battery, usually the power unit of the phone





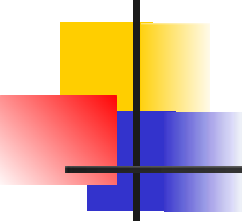
Subscriber Identity Module (SIM) card

- SIM – a memory card (integrated circuit) holding identity information, phone book etc.
- GSM system support SIM cards
- other systems, like CDMA do not support SIM cards, but have something similar called Re-Usable Identification Module (RUIM)



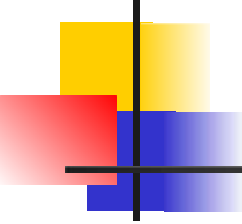
International Mobile Equipment Identity (IMEI) key

- IMEI – a unique 15 digit number identifying each phone, is incorporated in the cellular phone by the manufacturer
 - IMEI ex.: 994456245689001
 - when a phone tries to access a network, the service provider verifies its IMEI with a database of stolen phone numbers; if it is found in the database, the service provider denies the connection
- the IMEI is located on a white sticker/label under the battery, but it can also be displayed by typing *#06# on the phone



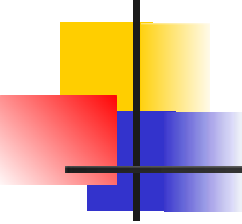
IMEI

- A 3-digit Type Approval Code (TAC). This is given to the unit after it passes conformance tests.
- A 1 or 2-digit Final Assembly Code (FAC). This identifies the place of final manufacture or assembly of the MS unit.
- The MS unit serial number.
- 1 spare digit reserved for future assignment
-



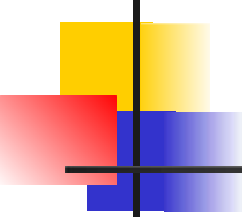
International Mobile Subscriber Identity (IMSI) key

- IMSI – a 15-digit unique number provided by the service provider and incorporated in the SIM card which identifies the subscriber
- IMSI enables a service provider to link a phone number with a subscriber



IMSI

- A 3-digit Mobile Country Code (MCC). This identifies the country where the GSM system operates.
- A 2-digit Mobile Network Code (MNC). This uniquely identifies each cellular provider.
- The Mobile Subscriber Identification Code (MSIC). This uniquely identifies each customer of the provider.



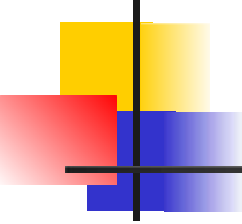
IMEI and IMSI structure

TAC (3 digits)	FAC (1 or 2 digits)	Serial number (up to 11 digits)	1 spare digit
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IMEI structure

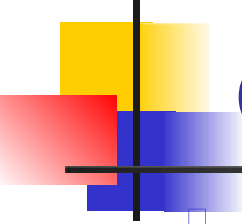
MCC (3 digits)	MNC (2 digits)	MSIC (up to 10 digits)
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IMSI structure



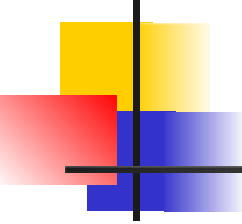
Temporary Mobile Subscriber Identity (TMSI) key

- TMSI – is a temporary number, shorter than the IMSI, assigned by the service provider to the phone on a temporary basis
- TMSI key identifies the phone and its owner in the cell it is located; when the phone moves to a different cell it gets a new TMSI key
- as TMSI keys are shorter than IMSI keys they are more efficient to send
- TMSI key are used for securing GSM networks



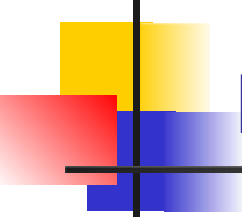
Cellular network components

- BTS (Base Transceiver Station) – main component of a cell and it connects the subscribers to the cellular network; for transmission/reception of information it uses several antennas spread across the cell
- BSC (Basic Station Controller) – it is an interface between BTSs and it is linked to BTSs by cable or microwave links; it routes calls between BTSs; it is also connected to the MSC
- MSC (Mobile Switching Center) – the coordinator of a cellular network, it is connected to several BSCs, it routes calls between BSCs; links the cellular network with other networks like PSTN through fiber optics, microwave or copper cable



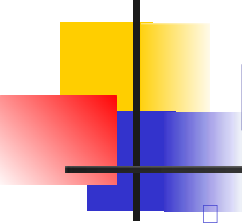
Base Station Subsystem

- Transcoding Rate and Adaptation Unit (TRAU)
 - Performs coding between the 64kbps PCM coding used in the backbone network and the 13 kbps coding used for the Mobile Station (MS)
- Base Station Controller (BSC)
 - Controls the channel (time slot) allocation implemented by the BTSes
 - Manages the handovers within BSS area
 - Knows which mobile stations are within the cell and informs the MSC/VLR about this
- Base Transceiver System (BTS)
 - Controls several transmitters
 - Each transmitter has 8 time slots, some used for signaling, on a specific frequency



Network and Switching Subsystem

- The backbone of a GSM network is a telephone network with additional cellular network capabilities
 - Mobile Switching Center (MSC)
 - Visitor Location Register (VLR)
 - Gateway Mobile Switching Center (GMSC)
 - Links the system to PSTN and other operators
 - Home Location Register (HLR)
 - Equipment Identity Register (EIR)

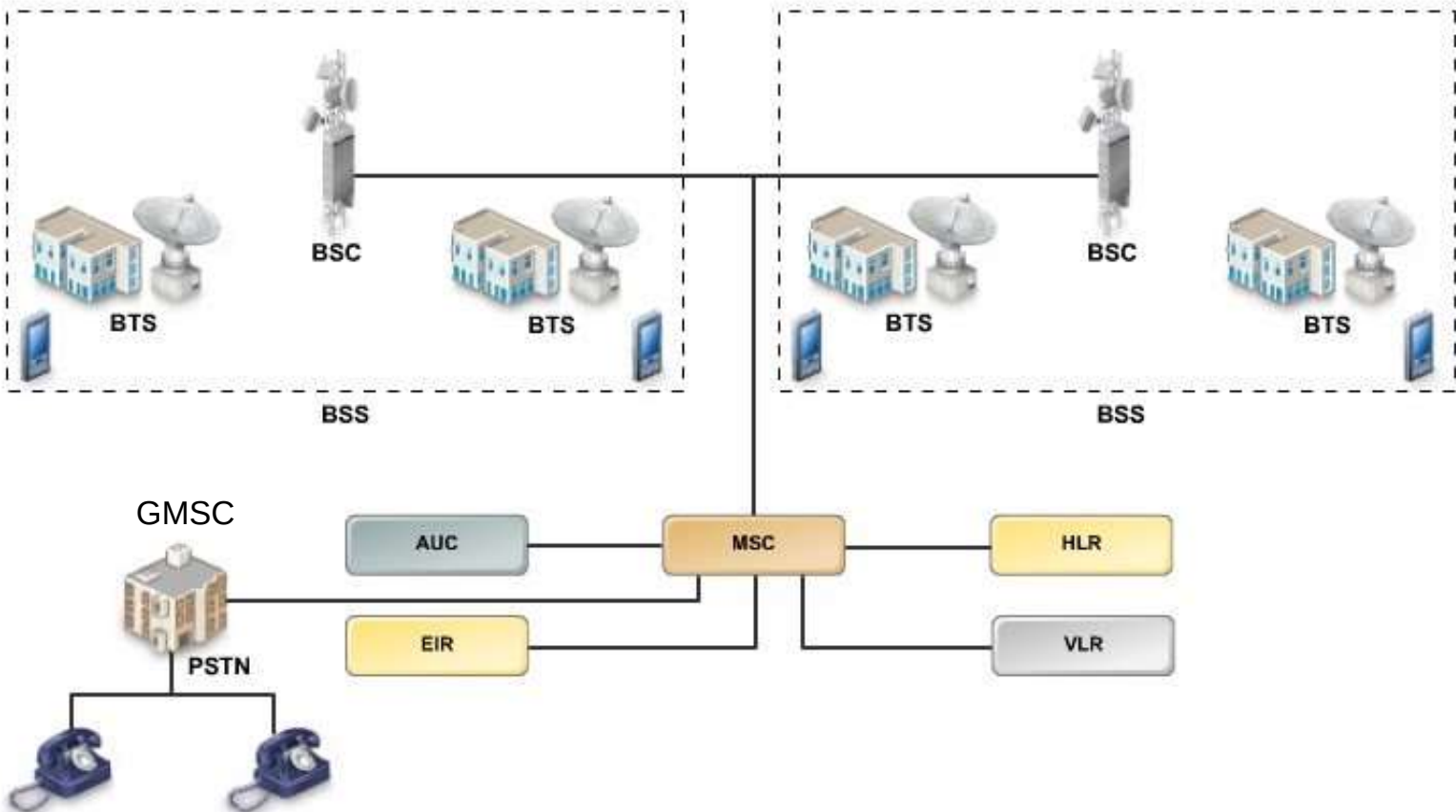


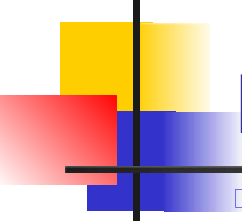
HLR and VLR

- Home Location Register (HLR) -
 - is a database maintained by the service provider containing permanent data about each subscriber (i.e. location, activity status, account status, call forwarding preference, caller identification preference, Contain subscriber information, authentication information in Authentication Center (AuC))
- Visitor Location Register (VLR) –
 - Contains the location of the active Mobile Stations
 - database that stores temporary data about a subscriber; it is kept in the MSC of the of the area the subscriber is located in; when the subscriber moves to a new area the new MSC requests this VLR from the HLR of the old MSC

GSM Mobile Switching Center (MSC)

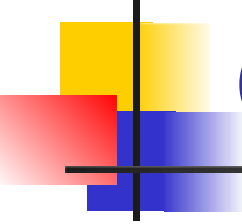
- is a switching center of the GSM network; coordinates BSCs linked to it





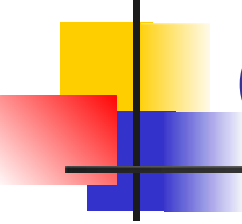
Mobile Switching Center

- The MSC (mobile switching center) plays a central role in GSM
 - switching functions
 - additional functions for mobility support management of network resources
 - interworking functions via Gateway MSC (GMSC) integration of several databases
 - specific functions for paging and call forwarding termination of SS7 (signaling system no. 7) mobility specific signaling
 - location registration and forwarding of location information provision of new services (fax, data calls)
 - support of short message service (SMS)
 - generation and forwarding of accounting and billing information
 -



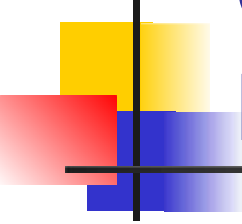
Operation subsystem

- The OSS (Operation Subsystem) enables centralized operation, management, and maintenance of all GSM subsystems Components
- - Authentication Center (AUC)
 - generates user specific authentication parameters on request of a VLR
 - authentication parameters used for authentication of mobile terminals and encryption of user data on the air interface within the GSM system
 - Equipment Identity Register (EIR)
 - registers GSM mobile stations and user rights
 - stolen or malfunctioning mobile stations can be locked and sometimes even localized
 - Operation and Maintenance Center (OMC)
 - different control capabilities for the radio subsystem and the network subsystem



GSM Services

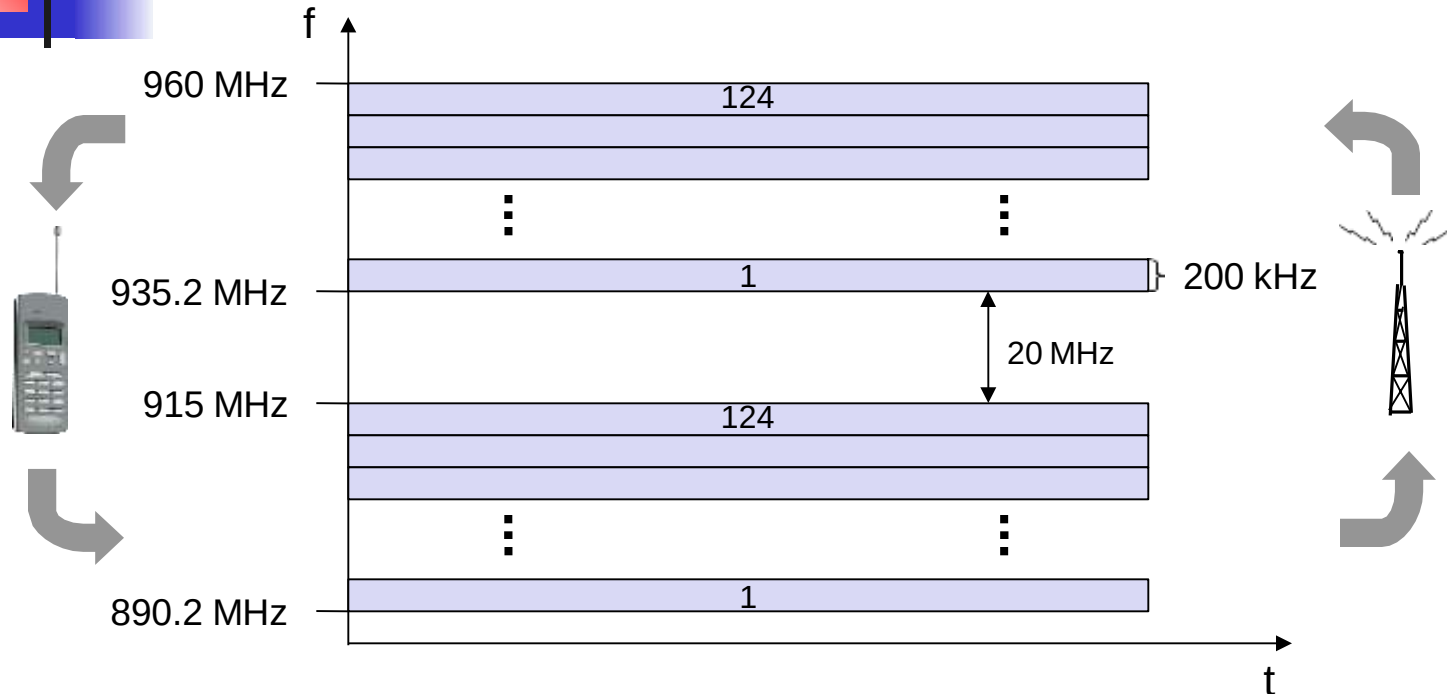
- Voice, 3.1 kHz
- Short Message Service (SMS)
 - 1985 GSM standard that allows messages of at most 160 chars. (incl. spaces) to be sent between handsets and other stations
 - Over 2.4 *billion* people use it; multi-billion \$ industry
- General Packet Radio Service (GPRS)
 - GSM upgrade that provides IP-based packet data transmission up to 114 kbps
 - Users can “simultaneously” make calls and send data
 - GPRS provides “always on” Internet access and the Multimedia Messaging Service (MMS) whereby users can send rich text, audio, video messages to each other
 - Performance degrades as number of users increase
 - GPRS is an example of 2.5G telephony – 2G service similar to 3G

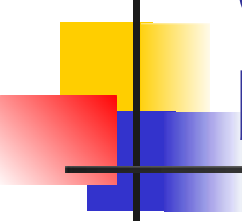


GSM Radio Subsystem and Frame Structure

- GSM uses FDMA and TDMA to transmit voice and data
 - the uplink channel between the cell phone and the BTS uses FDMA and a specific frequency band
 - the downlink channel between the BTS and the cell phone uses a different frequency band and the TDMA technique
 - there is sufficient frequency separation between the uplink freq. band and the downlink freq. band to avoid interference
- each uplink and downlink frequency bands is further split up as
- Control Channel (used to set up and manage calls) and Traffic Channel (used to carry voice)

FDD/FDMA - general scheme, example GSM

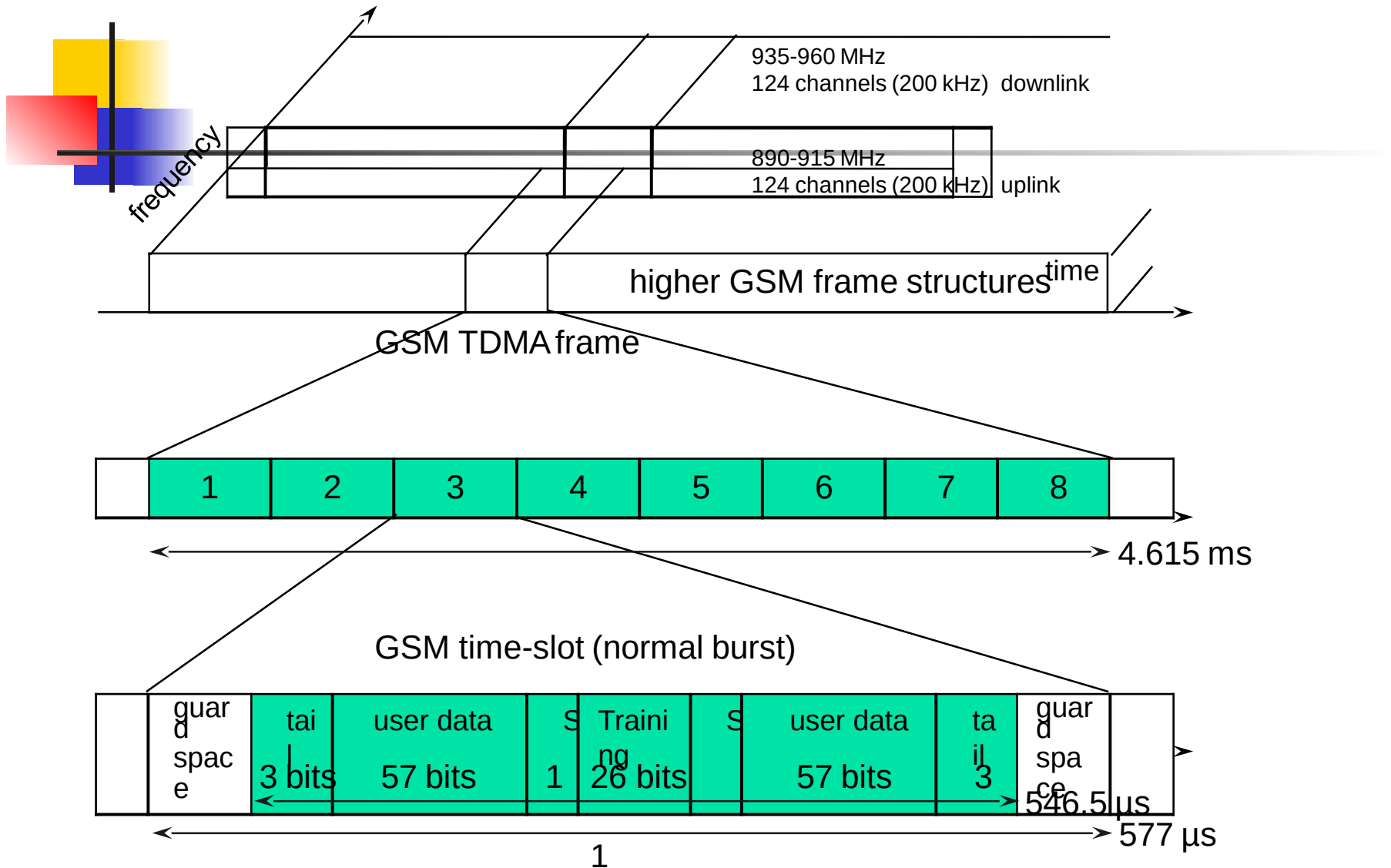




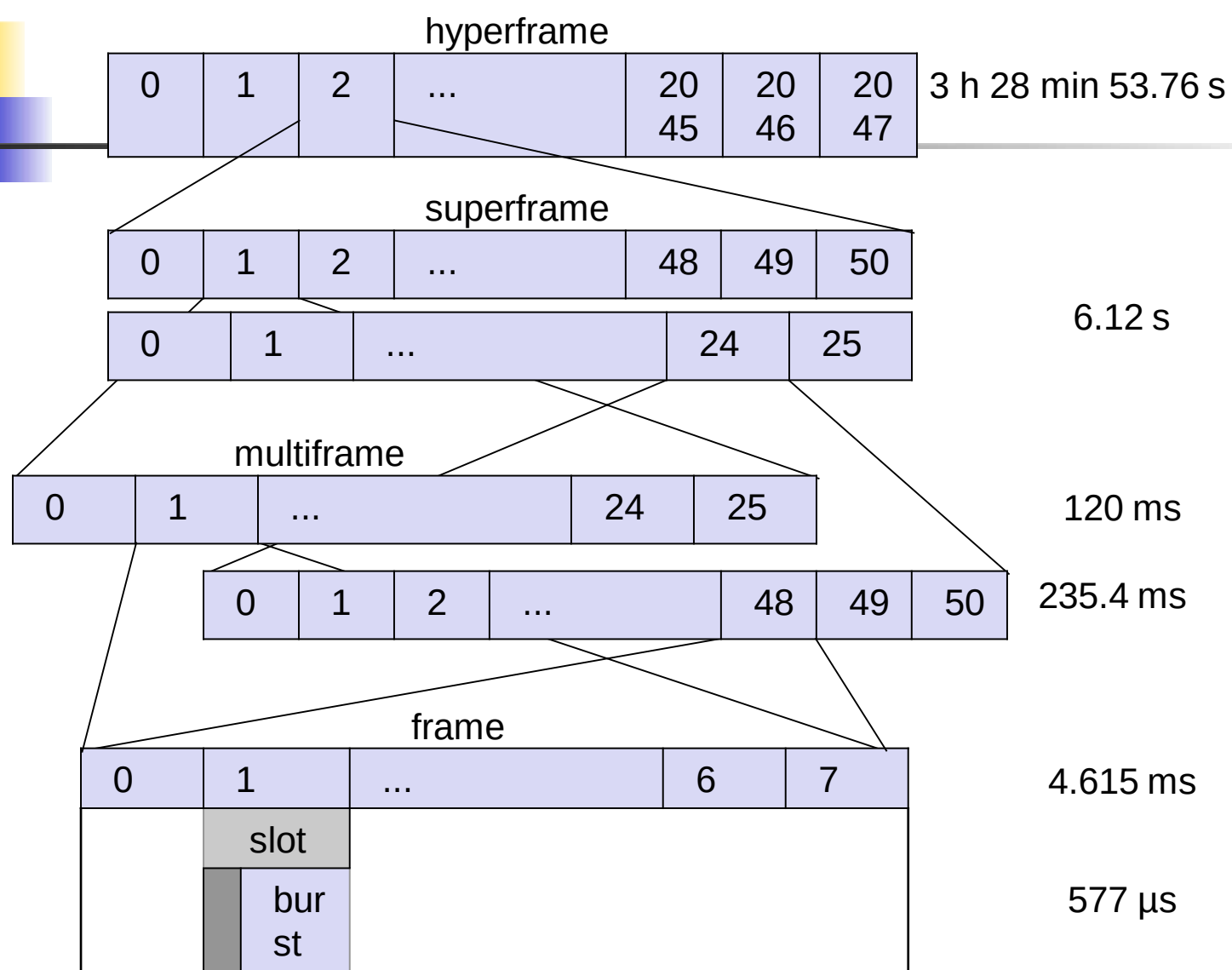
GSM uplink/downlink frequency bands

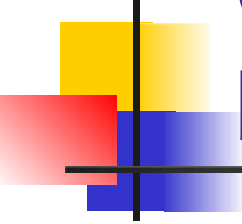
- uplink and downlink take place in different time slots using TDMA
- uplink and downlink channels have a bandwidth of 25 MHz
- these channels are further split up in a 124 carrier frequencies (1 control channels and the rest as traffic channels); each carrier frequency is spaced 200 KHz apart to avoid interference
- these carrier frequencies are further divided by time using TDMA and each time slot lasts for 0.577 ms.

GSM - TDMA/FDMA



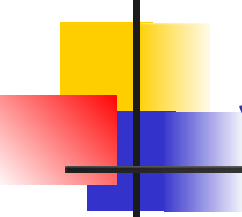
GSM hierarchy of frames





GSM uplink/downlink frequency bands used

GSM Frequency band	Uplink/BTS Transmit	Downlink/BTS Receive
900 MHz	935-960 MHz	890-915 MHz
1800 MHz	1805-1880 MHz	1710-1785 MHz
1900 MHz	1930-1990 MHz	1850-1910 MHz



Security in GSM

- Security services

- access control/authentication

- user □ SIM (Subscriber Identity Module): secret PIN (personal identification number)
 - SIM □ network: challenge response method

- confidentiality

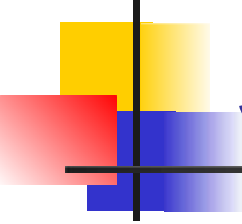
- voice and signaling encrypted on the wireless link (after successful authentication)

- anonymity

- temporary identity TMSI (Temporary Mobile Subscriber Identity)
 - newly assigned at each new location update
 - encrypted transmission

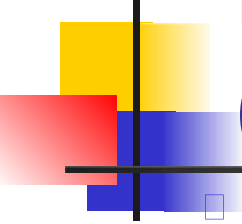
“secret”:

- A3 and A8 available via the Internet
- network providers can use stronger mechanisms



Security in GSM

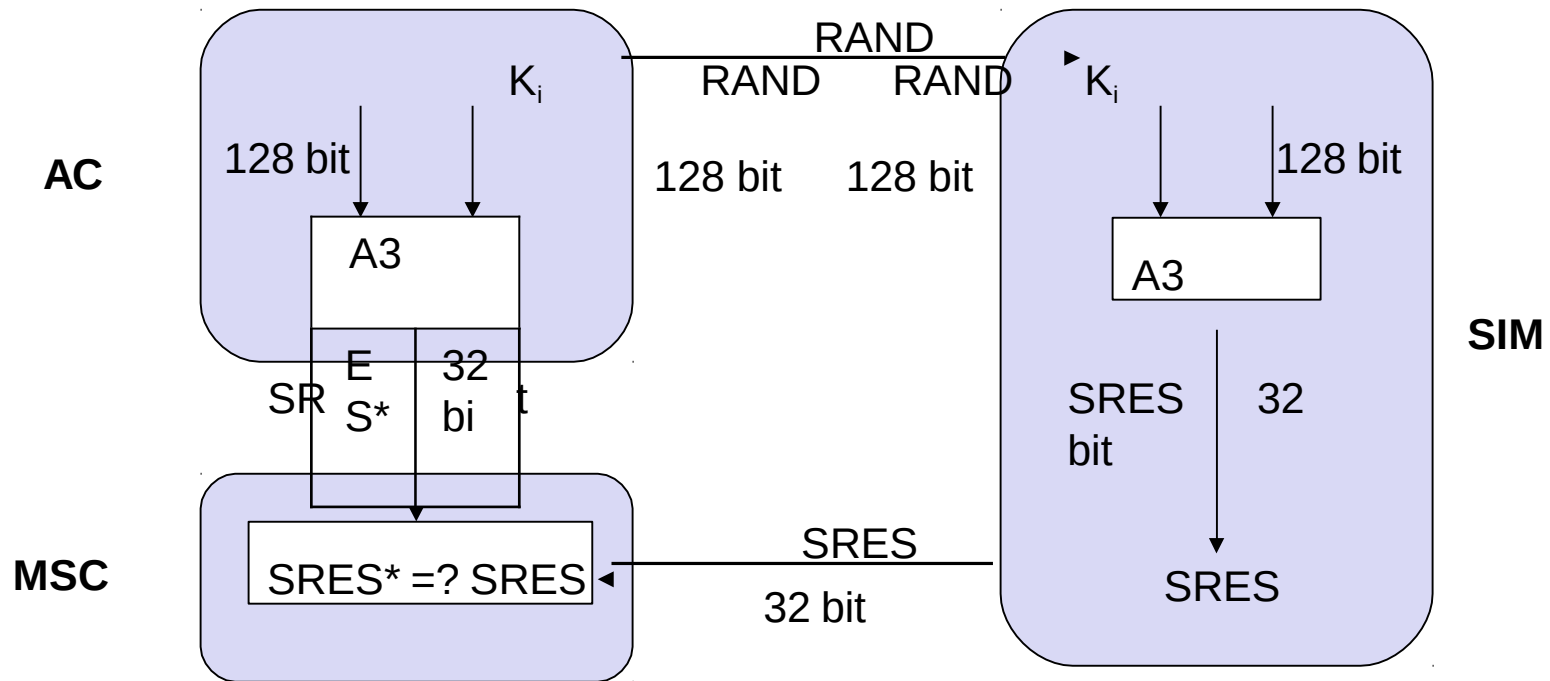
- 3 algorithms specified in GSM
 - A3 for authentication (“secret”, open interface) A5 for
 - encryption (standardized)
 - A8 for key generation (“secret”, open interface)



Personal Identification Number (PIN)

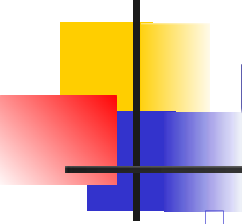
- the PIN is stored on the SIM card of the cell phone
- when the cell phone is turned on, the SIM checks the PIN; in case of 3 consecutive faulty PIN inputs a PUK (Personal Unblocking Key) is asked for
- in case of 10 faulty PUK inputs, the SIM is locked and the subscriber must ask a new SIM
- this security measure is within the cell phone and the service provider is not involved

GSM - authentication



K_i : individual subscriber authentication key

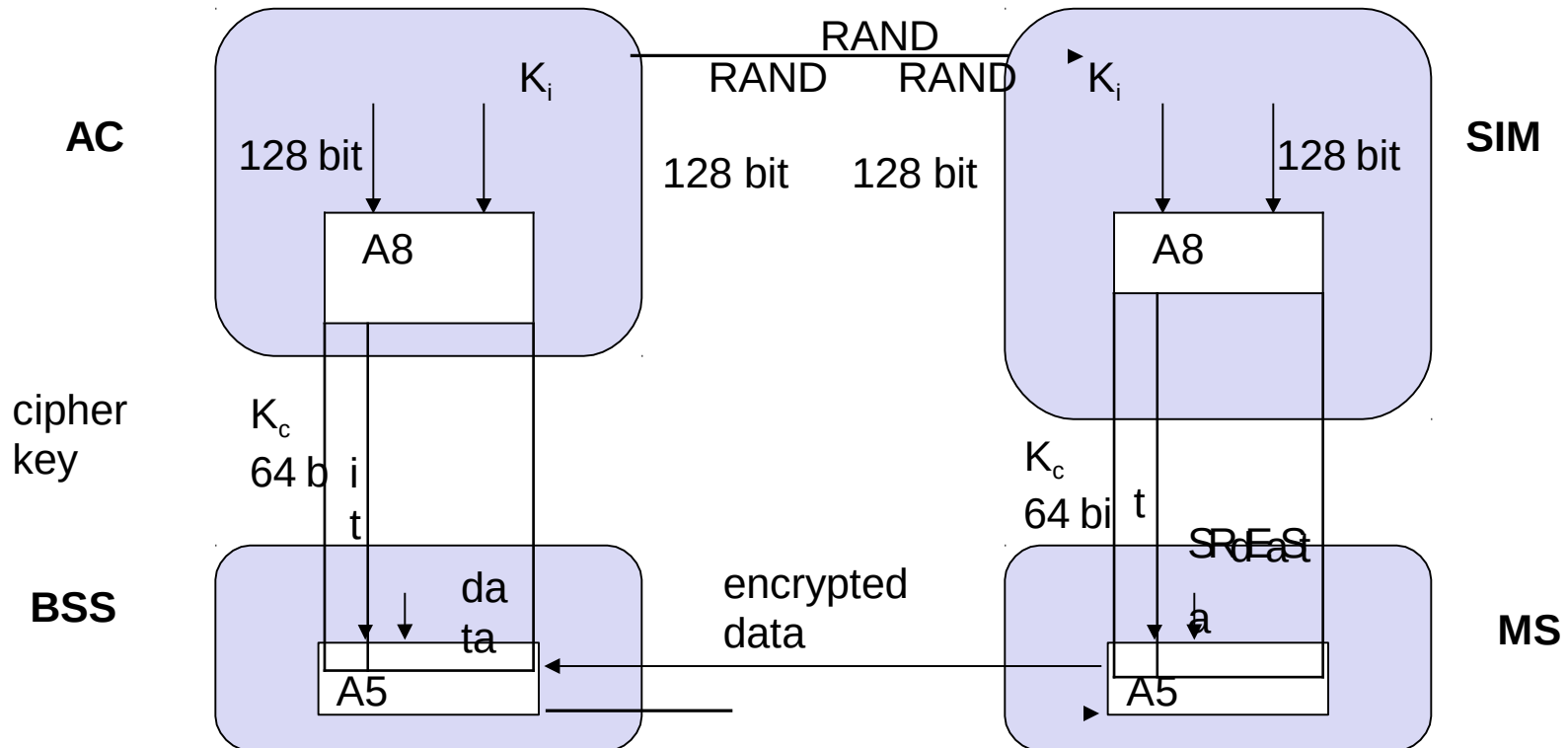
$SRES$: signed response

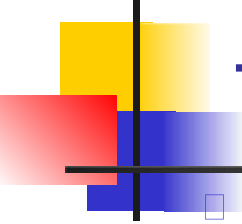


User Authentication

- a mechanism for encrypting messages in a GSM network
- the network sends random data to the cell phone (RAND)
- each cell phone is allocated a secret key (KI)
- using RAND and KI and the A3 encryption algorithm the cell phone generates a signed result (SRES) which is then sent to the network
- a similar process takes place in the network which generates a signed result specific to the cell phone
- the network compares its SRES with the SRES generated by the phone and in case of a match the cell phone is connected to the network
-

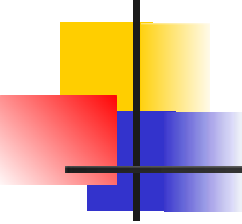
GSM - key generation and encryption





TMSI-Key Based Security

- is most used in a GSM cellular network
 - a TMSI key provides a temporary identification to a cell phone and is provided by the network upon authentication
 - a TMSI key keeps changing according to the location of the cell phone this way preventing unauthorized access to a channel and preventing intruder from tracing location
 - the mapping between IMSI and TMSI keys is handled by the VLR
 - IMSI are used only when the SIM is used for the first time



Roaming

- When a MS enters another operators network, it can be allowed to use the services of this operator
 - Operator to operator agreements and contracts
 - Higher billing
- The MS is identified by the information in the SIM card and the identification request is forwarded to the home operator
 - The home HLR is updated to reflect the